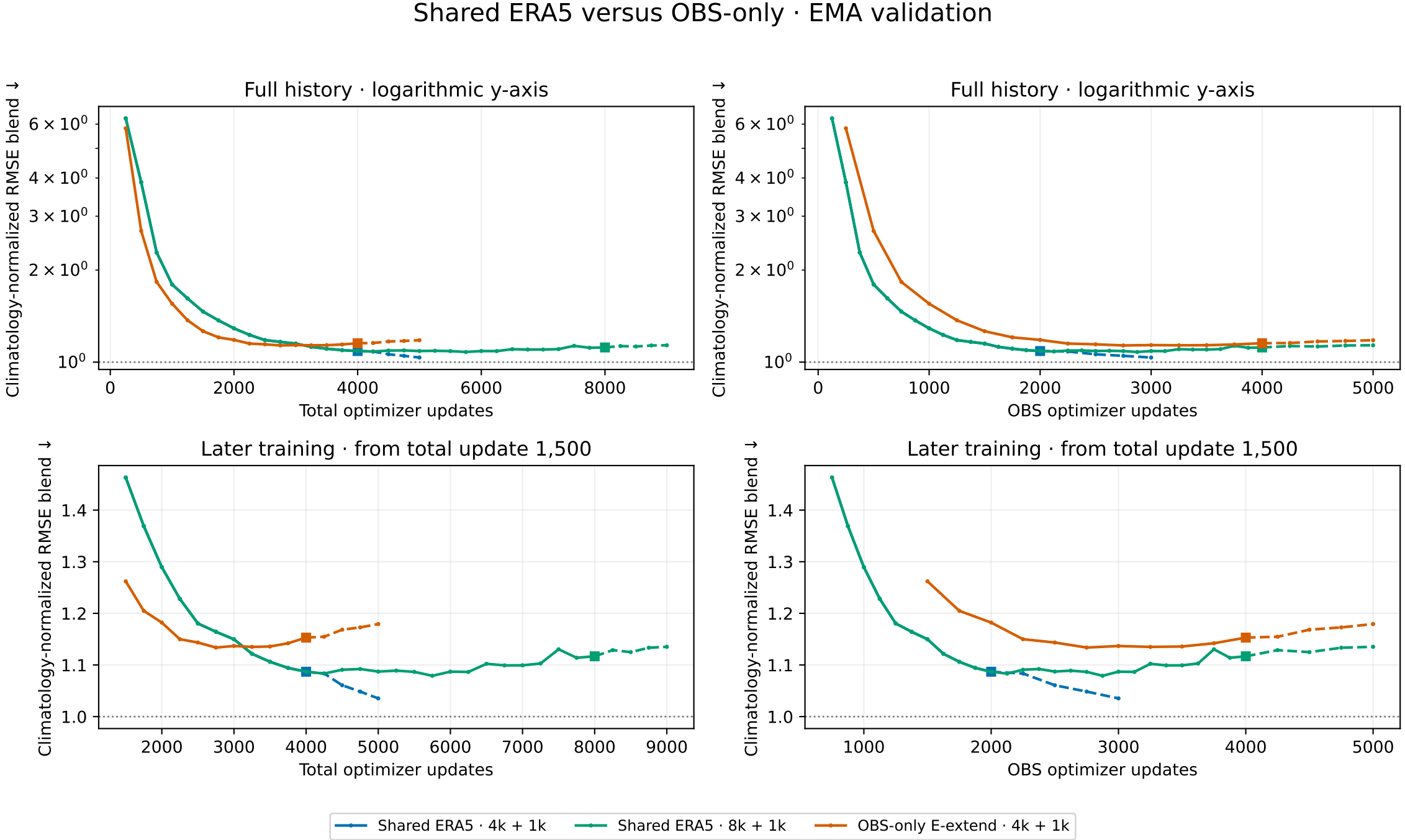
Shared ERA5 versus OBS-only · EMA validation
Full history · logarithmic y-axis
Climatology-normalized RMSE blend ↓
Total optimizer updates
OBS optimizer updates
Later training · from total update 1,500
Shared ERA5 · 4k + 1k
Shared ERA5 · 8k + 1k
OBS-only E-extend · 4k + 1k
Solid: main training. Dashed: OBS-only cooldown. Squares: cooldown start. One seed; identical nine-origin, eight-draw validation.